Final frozen browser-policy evaluation

6,000 unseen URLs • 5.00% phishing • no test tuning

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
HYB-MS real-only	97.95%	82.18%	75.33%	78.61%	95.87%	84.31%	0.86%	24.67%
HYB-MS +10% GAN	98.23%	84.15%	79.67%	81.85%	95.52%	87.18%	0.79%	20.33%
Browser ONNX v0.5.2	98.23%	84.15%	79.67%	81.85%	95.52%	87.18%	0.79%	20.33%

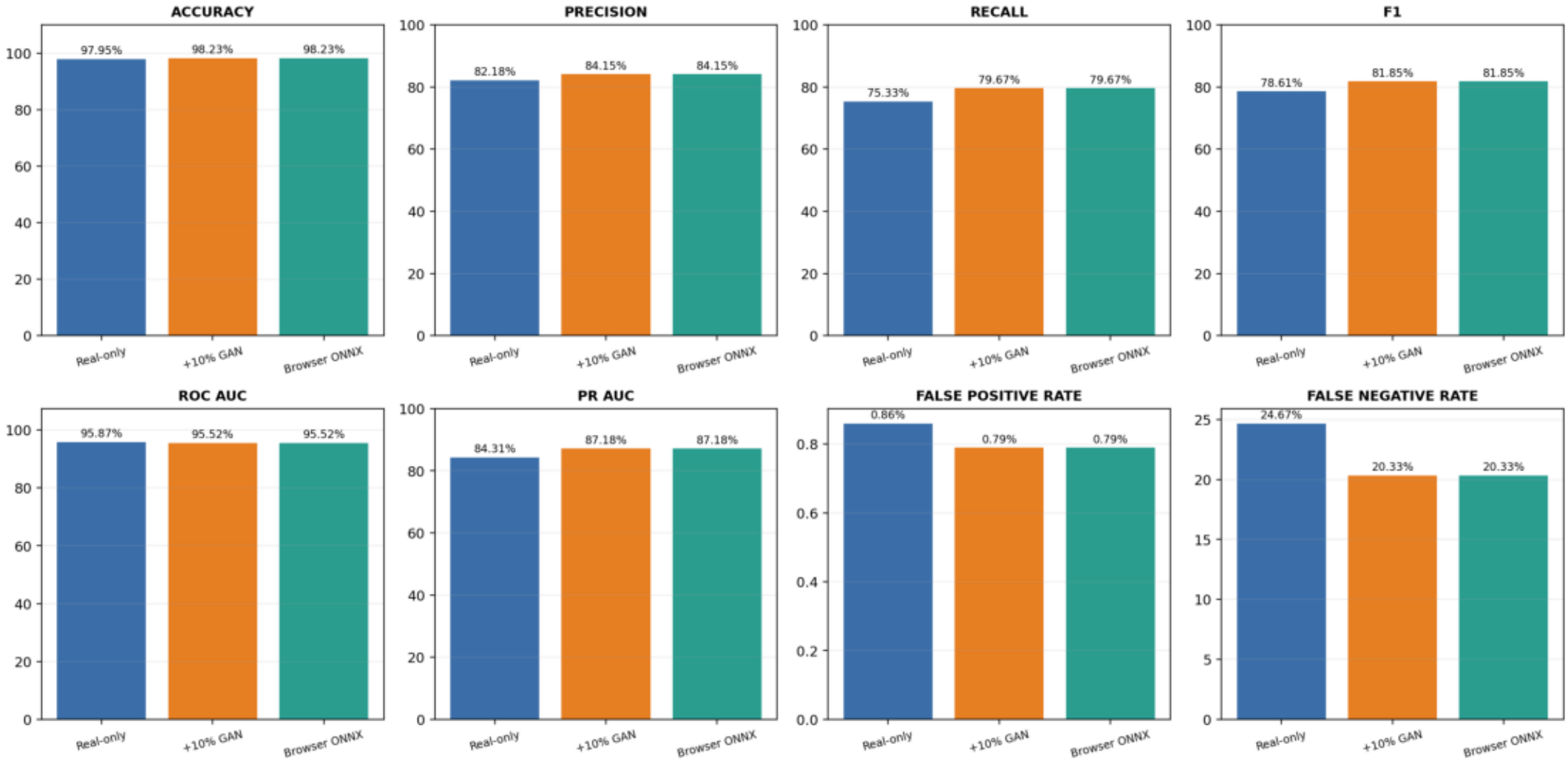
DEPLOYMENT GATE NOT PASSED

Neither native condition passed the pre-frozen FPR/recall deployment gate.

GAN augmentation: STRICT BENEFIT RULE NOT MET

The GAN condition produced a mixed or negative result under the rule frozen before final inference.

All thresholds were frozen on the separate policy-development split before this test file was opened.


Final frozen evaluation: model and browser comparison
ACCURACY
97.95%
98.23%
98.23%
PRECISION
82.18%
84.15%
84.15%
RECALL
75.33%
79.67%
79.67%
F1
78.61%
81.85%
81.85%
ROC AUC
95.87%
95.52%
95.52%
PR AUC
84.31%
87.18%
87.18%
FALSE POSITIVE RATE
0.86%
0.79%
0.79%
FALSE NEGATIVE RATE
24.67%
20.33%
20.33%
Real-only
+10% GAN
Browser ONNX

Error rates by important URL group

Group	Rows	Real error	GAN error	Browser error
Legitimate roots	5280	0.64%	0.51%	0.51%
Complex legit pages	420	3.57%	4.29%	4.29%
Legitimate login pages	25	4.00%	12.00%	12.00%
Long legitimate URLs	43	6.98%	6.98%	6.98%
All phishing	300	24.67%	20.33%	20.33%
Recent phishing	146	26.03%	17.81%	17.81%
Login-themed phishing	8	12.50%	12.50%	12.50%
Long phishing URLs	19	10.53%	10.53%	10.53%

For legitimate-only groups the error is FPR; for phishing-only groups it is FNR.